INL1, Problem 5

MAA155

EL OTMANI Youssef

$\star\ f = X^2Y^2 - XY + 2X^2Y - 2Y^2 - 2XY + X^2 - 4Y - X - 2$

$= (X^2 - X - 2)Y^2 + (2X^2 - 2X - 4)Y + (X^2 - X - 2)$

$f = (X^2 - X - 2)(Y^2 + 2Y + 1) = (X^2 - X - 2)(Y+1)^2$

So $c(f) = (X^2 - X - 2)$ and $pp(f) = (Y+1)^2$

$\star\ g = XY^2 + X^2Y + Y^2 + 2XY + X^2Y + X^2 + Y + X$

$= (X+1)Y^2 + (X^2 + 2X + 1)Y + (X^2 + X)$

$= (X+1)Y^2 + (X+1)^2Y + X(X+1) = (X+1)\left(Y^2 + (X+1)Y + X\right)$

So $c(g) = (X+1)$ and $pp(g) = \left(Y^2 + (X+1)Y + X\right)$

$$\begin{array}{r|l} X^2 - X - 2 & X+1 \\ \hline -\ X^2 + X & X - 2 \\ \hline 0 - 2X - 2 & \\ -\ -2X - 2 & \\ \hline 0 & \end{array}$$

$$\begin{array}{r|l} Y^2 + Y + XY + X & Y+1 \\ \hline -\ Y^2 + Y & Y + X \\ \hline 0 + XY + X & \\ -\ XY + X & \\ \hline 0 & \end{array}$$

So $\gcd(c(f), c(g)) = \gcd\left((X-2)(X+1), X+1\right) = X+1$

And $\gcd(pp(f), pp(g)) = \gcd\left((Y+1)^2, (Y+1)(Y+X)\right) = Y+1$

So $\gcd(f, g) = (X+1)(Y+1) = XY + X + Y + 1$

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#EL OTMAI Youssef
Ring.<x, y> = QQ[]
A= (x^2)*(y^2) - x * (y^2) + 2 * (x^2) * y - 2 * (y^2) - 2 * x * y + x^2 - 4*y - x - 2
B = x*(y^2) + (x^2) * y + (y^2) + 2 * x * y + x^2 + y + x
gcd(A,B)
Out :
x*y + x + y + 1
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